

“Scheduling”

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Learning Outcome

- ❖ At the end of this lecture the students should be able to explain scheduling, priority rules of scheduling
- ❖ At the end of this lecture the students should be able to describe types of scheduling
- ❖ At the end of this lecture the students should be able to calculate different priority rules of scheduling
- ❖ At the end of this lecture the students should be able to solve Johnson's Rule

Scheduling

Scheduling means Establishing the timing of the use of equipment, facilities, and human activities in an organization

Type of Scheduling

Job-shop scheduling

A work location in which a number of general purpose work stations exist and are used to perform a variety of jobs



Flow-shop scheduling

Flow-shop scheduling is a special case of job-shop scheduling where there is strict order of all operations to be performed on all jobs

Sequencing Jobs

- ☑ Specifies the order in which jobs should be performed at work centers
- ☑ Priority rules are used to dispatch or sequence jobs
 - ☑ FCFS: First come, first served
 - ☑ SPT: Shortest processing time
 - ☑ EDD: Earliest due date
 - ☑ LPT: Longest processing time

Priority Rules for Dispatching Jobs

First come, first served

The first job to arrive at a work center is processed first

Earliest due date

The job with the earliest due date is processed first

Shortest processing time

The job with the shortest processing time is processed first

Longest processing time

The job with the longest processing time is processed first

Sequencing Example

Apply the four popular sequencing rules to these five jobs

<i>Job</i>	<i>Job Work (Processing) Time (Days)</i>	<i>Job Due Date (Days)</i>
<i>A</i>	6	8
<i>B</i>	2	6
<i>C</i>	8	18
<i>D</i>	3	15
<i>E</i>	9	23

Sequencing Example

FCFS: Sequence A-B-C-D-E

<i>Job Sequence</i>	<i>Job Work (Processing) Time</i>	<i>Flow Time</i>	<i>Job Due Date</i>	<i>Job Lateness</i>
<i>A</i>	6	6	8	0
<i>B</i>	2	8	6	2
<i>C</i>	8	16	18	0
<i>D</i>	3	19	15	4
<i>E</i>	9	28	23	5
	28	77		11

Sequencing Example

FCFS: Sequence A-B-C-D-E

$$\text{Average completion time} = \frac{\text{Sum of total flow time}}{\text{Number of jobs}} =$$

$$\text{Utilization} = \frac{\text{Total job work time}}{\text{Sum of total flow time}} =$$

$$\text{Average number of jobs in the system} = \frac{\text{Sum of total flow time}}{\text{Total job work time}} =$$

$$\text{Average job lateness} = \frac{\text{Total late days}}{\text{Number of jobs}} =$$

Sequencing Example

FCFS: Sequence A-B-C-D-E

$$\text{Average completion time} = \frac{\text{Sum of total flow time}}{\text{Number of jobs}} = 77/5 = 15.4 \text{ days}$$

$$\text{Utilization} = \frac{\text{Total job work time}}{\text{Sum of total flow time}} = 28/77 = 36.4\%$$

$$\text{Average number of jobs in the system} = \frac{\text{Sum of total flow time}}{\text{Total job work time}} = 77/28 = 2.75 \text{ jobs}$$

$$\text{Average job lateness} = \frac{\text{Total late days}}{\text{Number of jobs}} = 11/5 = 2.2 \text{ days}$$

Sequencing Example

SPT: Sequence B-D-A-C-E

<i>Job Sequence</i>	<i>Job Work (Processing) Time</i>	<i>Flow Time</i>	<i>Job Due Date</i>	<i>Job Lateness</i>
<i>B</i>	2	2	6	0
<i>D</i>	3	5	15	0
<i>A</i>	6	11	8	3
<i>C</i>	8	19	18	1
<i>E</i>	9	28	23	5
	28	65		9

Sequencing Example

SPT: Sequence B-D-A-C-E

$$\text{Average completion time} = \frac{\text{Sum of total flow time}}{\text{Number of jobs}} = 65/5 = 13 \text{ days}$$

$$\text{Utilization} = \frac{\text{Total job work time}}{\text{Sum of total flow time}} = 28/65 = 43.1\%$$

$$\text{Average number of jobs in the system} = \frac{\text{Sum of total flow time}}{\text{Total job work time}} = 65/28 = 2.32 \text{ jobs}$$

$$\text{Average job lateness} = \frac{\text{Total late days}}{\text{Number of jobs}} = 9/5 = 1.8 \text{ days}$$

Sequencing Example

EDD: Sequence B-A-D-C-E

<i>Job Sequence</i>	<i>Job Work (Processing) Time</i>	<i>Flow Time</i>	<i>Job Due Date</i>	<i>Job Lateness</i>
<i>B</i>	2	2	6	0
<i>A</i>	6	8	8	0
<i>D</i>	3	11	15	0
<i>C</i>	8	19	18	1
<i>E</i>	9	28	23	5
	28	68		6

Sequencing Example

EDD: Sequence B-A-D-C-E

$$\text{Average completion time} = \frac{\text{Sum of total flow time}}{\text{Number of jobs}} = 68/5 = 13.6 \text{ days}$$

$$\text{Utilization} = \frac{\text{Total job work time}}{\text{Sum of total flow time}} = 28/68 = 41.2\%$$

$$\text{Average number of jobs in the system} = \frac{\text{Sum of total flow time}}{\text{Total job work time}} = 68/28 = 2.43 \text{ jobs}$$

$$\text{Average job lateness} = \frac{\text{Total late days}}{\text{Number of jobs}} = 6/5 = 1.2 \text{ days}$$

Sequencing Example

LPT: Sequence E-C-A-D-B

<i>Job Sequence</i>	<i>Job Work (Processing) Time</i>	<i>Flow Time</i>	<i>Job Due Date</i>	<i>Job Lateness</i>
<i>E</i>	9	9	23	0
<i>C</i>	8	17	18	0
<i>A</i>	6	23	8	15
<i>D</i>	3	26	15	11
<i>B</i>	2	28	6	22
	28	103		48

Sequencing Example

LPT: Sequence E-C-A-D-B

$$\text{Average completion time} = \frac{\text{Sum of total flow time}}{\text{Number of jobs}} = 103/5 = 20.6 \text{ days}$$

$$\text{Utilization} = \frac{\text{Total job work time}}{\text{Sum of total flow time}} = 28/103 = 27.2\%$$

$$\text{Average number of jobs in the system} = \frac{\text{Sum of total flow time}}{\text{Total job work time}} = 103/28 = 3.68 \text{ jobs}$$

$$\text{Average job lateness} = \frac{\text{Total late days}}{\text{Number of jobs}} = 48/5 = 9.6 \text{ days}$$

Sequencing Example

Summary of Rules

<i>Rule</i>	<i>Average Completion Time (Days)</i>	<i>Utilization (%)</i>	<i>Average Number of Jobs in System</i>	<i>Average Lateness (Days)</i>
<i>FCFS</i>	15.4	36.4	2.75	2.2
<i>SPT</i>	13.0	43.1	2.32	1.8
<i>EDD</i>	13.6	41.2	2.43	1.2
<i>LPT</i>	20.6	27.2	3.68	9.6

Comparison of Sequencing Rules

- ✓ No one sequencing rule excels on all criteria
- ✓ SPT does well on minimizing flow time and number of jobs in the system
- ✓ But SPT moves long jobs to the end which may result in dissatisfied customers
- ✓ FCFS does not do especially well (or poorly) on any criteria but is perceived as fair by customers
- ✓ EDD minimizes lateness

Sequencing N Jobs on Two Machines: Johnson's Rule

- ☑ Works with two or more jobs that pass through the same two machines or work centers
- ☑ Minimizes total production time and idle time

Johnson's Rule

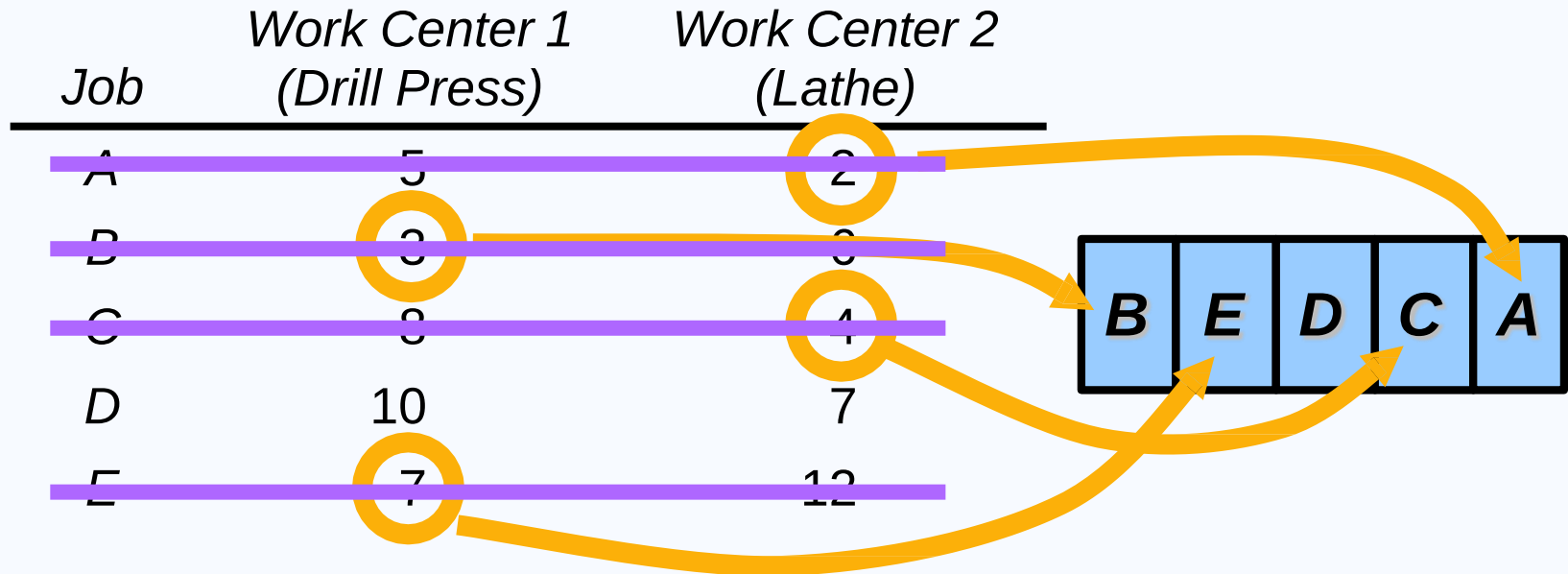
1. List all jobs and times for each work center
2. Choose the job with the shortest activity time. If that time is in the first work center, schedule the job first. If it is in the second work center, schedule the job last.
3. Once a job is scheduled, it is eliminated from the list
4. Repeat steps 2 and 3 working toward the center of the sequence

Johnson's Rule Example

<i>Job</i>	<i>Work Center 1 (Drill Press)</i>	<i>Work Center 2 (Lathe)</i>
<i>A</i>	5	2
<i>B</i>	3	6
<i>C</i>	8	4
<i>D</i>	10	7
<i>E</i>	7	12

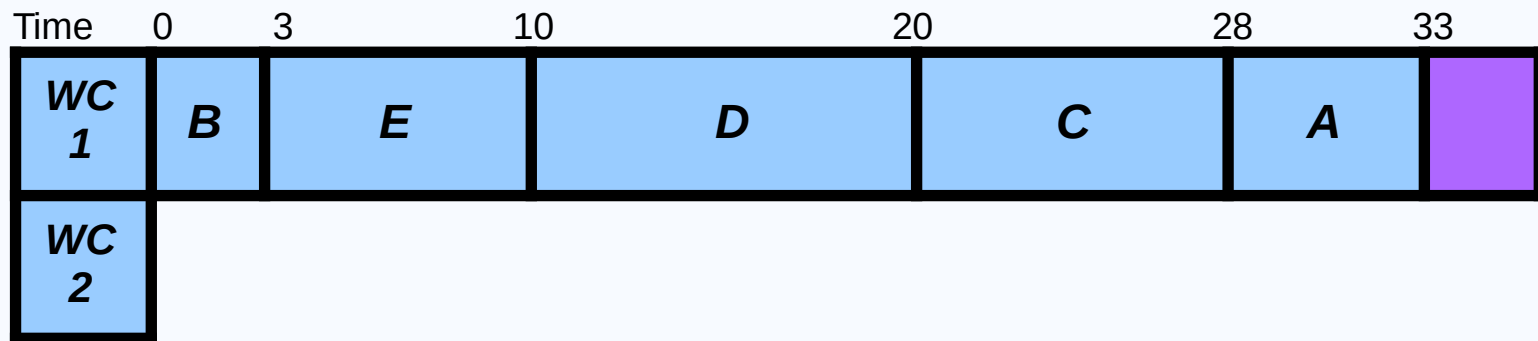
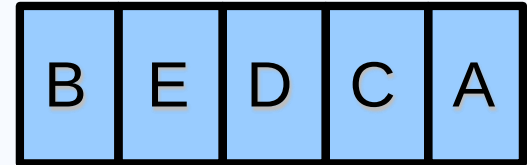


Johnson's Rule Example



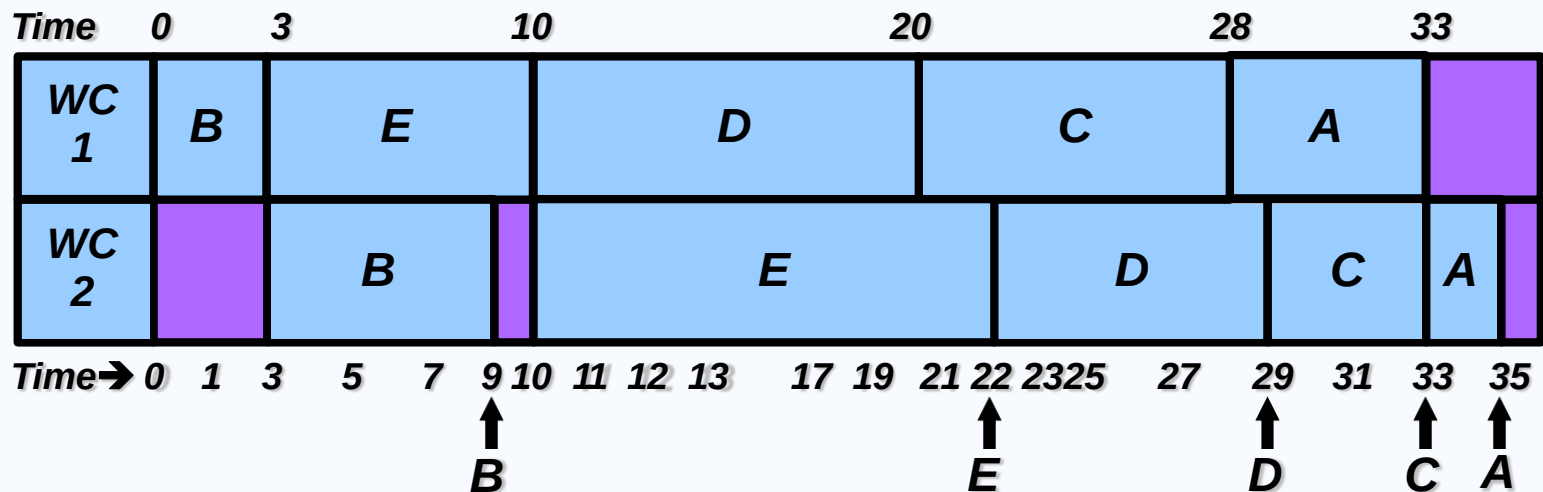
Johnson's Rule Example

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<i>D</i>	10	7
<i>E</i>	7	12



Johnson's Rule Example

Job	Work Center 1 (Drill Press)	Work Center 2 (Lathe)
A	5	2
B	3	6
C	8	4
D	10	7
E	7	12



**THANK
YOU !**